

Coding & Solution

Date	• 14 July 2026
Team ID	• XXXXXX
Project Name	AgroSense AI – Intelligent Crop Recommendation System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field

Details

Repository Link / URL	https://crop-reco.vercel.app/
Programming Lang	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• Crop recommendation using Machine Learning. • Accepts N, P, K, Temperature, Humidity, pH, and Rainfall as inputs. • Validates user input.
Pending / Incomplete Features	• Weather API integration. • User authentication and profile management. • Recommendation history.
Setup / Run Instructions	• Prediction accuracy depends on the training dataset. • Requires valid soil and weather parameter values.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular Flask architecture with separate frontend and backend components. Meaningful naming conventions, input validation, and a machine learning prediction module improve readability and maintainability. The source code is version-controlled using GitHub, and the application is successfully deployed on Vercel. Future enhancements include weather API integration, user authentication, and prediction history.